

Chengsong Huang

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EDUCATION

Washington University in St. Louis

Advisor: Prof. Jiaxin Huang

Ph.D. in Computer Science (GPA 4.0/4.0)

Sep. 2023 - Present

St. Louis, MO

Fudan University

Software School of Fudan University

B.S. in Software Engineering (GPA 3.82/4.0) rank 1%

Sep. 2019 - Jun. 2023

Shanghai, China

RESEARCH INTERESTS

My research focuses on advancing the capabilities and accessibility of Large Language Models (LLMs). My interests include:

- **Efficient LLMs:** Developing and analyzing methods to make LLMs more computationally efficient. My experience includes work on parameter-efficient fine-tuning (PEFT) and speculative decoding to accelerate inference speed.
- **Self-Improving Language Models:** Designing frameworks that enable LLMs to iteratively refine their own outputs. This includes research on self-correction and learning from machine-generated feedback to autonomously enhance model performance.

PREPRINTS (SELECTED)

* stands for equal contribution or alphabetical ordering.

- **Huang, C.**, Liu, H., Zheng, T., Dai, R., Huang, L., Li, J., Li, Z., Wei, Z., Meng, Y. and Huang, J. “[G-Zero: Self-Play for Open-Ended Generation from Zero Data](#)”. arXiv preprint. 2026.
- Huang, L., **Huang, C.**, Li, J., Cai, D., Yang, Y. and Huang, J. “[Nonsense Helps: Prompt Space Perturbation Broadens Reasoning Exploration](#)”. arXiv preprint. 2026.
- **Huang, C.**, Zheng, T., Huang, L., Li, J., Liu, H. and Huang, J. “[RelayLLM: Efficient Reasoning via Collaborative Decoding](#)”. arXiv preprint. 2026.
- Qian, Q*, **Huang, C.***, Xu, J., Lv, C., Wu, M., Liu, W., Wang, X., Wang, Z., Huang, Z., Tian, M., Xu, J., Hu, K., Wang, H.-D., Hu, Y., Huang, X. and Zheng, X. “[Benchmark²: Systematic Evaluation of LLM Benchmarks](#)”. arXiv preprint. 2026.
- Tang, Y., **Huang, C.**, Huang, J. and Yeoh, W. “[UniRel-R1: RL-tuned LLM Reasoning for Knowledge Graph Relational Question Answering](#)”. arXiv preprint. 2025.
- Huang, L., **Huang, C.**, Leng, J., Huang, D. and Huang, J. “[POSS: Position Specialist Generates Better Draft for Speculative Decoding](#)”. arXiv preprint. 2025.
- Yu, W., Liang, Z., **Huang, C.**, Panaganti, K., Fang, T., Mi, H. and Yu, D. “[Guided Self-Evolving LLMs with Minimal Human Supervision](#)”. arXiv preprint. 2025.

PUBLISHED PAPERS

For a complete list of publications, please visit my [Google Scholar profile](#).

* stands for equal contribution or alphabetical ordering.

- Zheng, T.*, **Huang, C.***, Dai, R.*, He, Y., Liu, R., Ni, X., Bao, H., Wang, K., Zhu, H., Huang, J., Huang, F. and Huang, H. “[Parallel-Probe: Towards Efficient Parallel Thinking via 2D Probing](#)”. ICML 2026.
- Li, J., **Huang, C.**, Huang, L., Xu, S., Liu, H., Zhang, W. and Huang, J. “[Training Data Efficiency in Multimodal Process Reward Models](#)”. ICML 2026.
- **Huang, C.**, Yu, W., Wang, X., Zhang, H., Li, Z., Li, R., Huang, J., Mi, H. and Yu, D. “[R-Zero: Self-Evolving Reasoning LLM from Zero Data](#)”. ICLR 2026.
- **Huang, C.**, Huang, L., Leng, J., Liu, J. and Huang, J. “[Efficient Test-Time Scaling via Self-Calibration](#)”. ICLR 2026.
- He, Y.*, **Huang, C.***, Li, Z.*, Huang, J. and Yang, Y. “[VisPlay: Self-Evolving Vision-Language Models from Images](#)”. CVPR 2026.
- **Huang, C.**, Huang, L. and Huang, J. “[Divide, Reweigh, and Conquer: a Logit Arithmetic Approach for In-Context Learning](#)”. EACL 2026.
- Leng, J., **Huang, C.**, Huang, L., Lin, B. Y., Cohen, W. W., Wang, H. and Huang, J. “[CrossWordBench: Evaluating the Reasoning Capabilities of LLMs and LVLMs with Controllable Puzzle Generation](#)”. COLM 2025.
- Kong, L., Feng, J., Liu, H., **Huang, C.**, Huang, J., Chen, Y. and Zhang, M. “[GOFA: A Generative One-For-All Model for Joint Graph Language Modeling](#)”. ICLR 2025.
- Leng, J., **Huang, C.**, Zhu, B. and Huang, J. “[Taming Overconfidence in LLMS: Reward Calibration in RLHF](#)”. ICLR 2025.
- **Huang, C.***, Liu, Q.*, Lin, B. Y.*, Pang, T., Du, C. and Lin, M. “[LoraHub: Efficient Cross-Task Generalization via Dynamic LoRA Composition](#)”. COLM 2024.
- Gu, C., Zheng, X., Xu, J., Wu, M., Zhang, C., **Huang, C.**, Cai, H. and Huang, X. “[Watermarking PLMs on Classification Tasks by Combining Contrastive Learning with Weight Perturbation](#)”. Findings of EMNLP 2023.
- Chen, L., **Huang, C.**, Zheng, X., Lin, J. and Huang, X. “[TableVLM: Multi-modal Pre-training for Table Structure Recognition](#)”. ACL 2023.
- Lin, B. Y.*, **Huang, C.***, Liu, Q., Gu, W., Sommerer, S. and Ren, X. “[On Grounded Planning for Embodied Tasks with Language Models](#)”. AAAI 2023.
- Xie, C., Huang, W., Liang, J., **Huang, C.** and Xiao, Y. “[WebKE: Knowledge Extraction from Semi-structured Web with Pre-trained Markup Language Model](#)”. CIKM 2021.
- Xie, C., Liang, J., Liu, J., **Huang, C.**, Huang, W. and Xiao, Y. “[Revisiting the Negative Data of Distantly Supervised Relation Extraction](#)”. ACL 2021.

EXPERIENCE

Google Cloud AI

Feb. 2026 - present

Student Researcher (Mentor: Zifeng Wang)

Agentic Reinforcement Learning for LLMs

Tencent AI Lab Seattle

Jun. 2025 - Aug. 2025

Research intern (Mentor: Wenhao Yu)

R-Zero: Self-Evolving Reasoning LLM from Zero Data

Sea AI lab (SAIL)

Jan. 2023 - Jun. 2023

Research intern (Mentor: Qian Liu)

LoraHub: Efficient Cross-Task Generalization via Dynamic LoRA Composition

USC INK lab

May. 2022 - Jan. 2023

Research intern (Mentor: Xiang Ren, Bill Yuchen Lin)

On Grounded Planning for Embodied Tasks with Language Models

SKILLS

- **Programming Languages:** Proficient in Python, SQL and Java.
- **Libraries & Frameworks:** Expertise in PyTorch, TensorFlow, Hugging Face Transformers, and Scikit-learn.
- **Tools & Platforms:** Proficient with Git and Wandb. Experience working with LSF and Slurm.

SERVICE

Conference Reviewer: ARR 2024-present, ICLR 2025-present, COLM 2024-present, AAAI 2026, CVPR 2026, ICML 2026.